Regulatory Applicability and Compliance-Gap Mapping of E-Waste (Management) Rules, 2022 for Leading Indian Electronics Manufacturers

Prepared for Academic Reference Purposes

Compiled with reference to the CPCB FAQ on E-Waste (Management) Rules, 2022, and Chadha, A. & Chadha, N. (2022). "Law Relating to E-Waste, its Impacts and Disposal Management." Neuro Quantology, 20(9), 1264-1273.

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Abstract

This report maps the obligations arising under India's E-Waste (Management) Rules, 2022 onto five leading Indian electronics manufacturers - Dixon Technologies, Kaynes Technology, Foxconn India, Tata Electronics and VVDN Technologies - identifying which of ten core regulatory provisions apply to each, based on their disclosed manufacturing role (Producer, Manufacturer, or Contract Manufacturer/EMS). It further sets out sector-wide compliance gaps documented in the e-waste law and policy literature - informal-sector dependence, opaque public recycler data, the collection-generation gap, weak penal enforcement on cross-border flows, and the absence of Life-Cycle-Assessment-based EPR design or consumer take-back incentives - as systemic risk areas relevant to the entire Indian EEE-producer base rather than as findings specific to any one company.

Keywords: *E-Waste (Management) Rules 2022, EPR, RoHS, Compliance Mapping, Electronics Manufacturers, India*

TABLE 1. CORE RULES CONSIDERED

No.	Rule / Obligation	Description
R1	**EPR Registration on the E-Waste EPR Portal**	Manufacturer/Producer registration on the CPCB E-Waste EPR Portal is mandatory before sale of EEE listed in Schedule-I.
R2	**Extended Producer Responsibility (EPR) Obligation**	Phased recycling targets - 60% (FY23-24 & 24-25), 70% (FY25-26 & 26-27), 80% (FY27-28 onward) of generation - fulfilled only through registered recyclers.
R3	**Purchase of EPR Certificates from Registered Recyclers**	EPR obligation is discharged by purchasing certificates (gold, copper, aluminium, iron end-products) from EPR-Portal-registered recyclers for the relevant EEE code.
R4	**Reduction of Hazardous Substances (RoHS) Compliance**	Under Rule 16(1), new EEE must not exceed 0.1% (0.01% for cadmium) by weight of Lead, Mercury, Cadmium, Hexavalent Chromium, PBB and PBDE in homogeneous materials.
R5	**RoHS Self-Declaration & Technical Documentation**	Self-declaration of RoHS compliance must be filed on the Portal; technical documents (EN 50581/EN IEC 63000:2018) kept available for CPCB/SPCB verification.
R6	**Quarterly and Annual Returns**	Sales, EPR obligation and fulfilment must be reported on the EPR Portal quarterly and annually.
R7	**Consent to Operate (CTO) / Hazardous Waste Authorisation**	Manufacturing facilities must hold valid CTO under the Air & Water Acts plus Hazardous and Other Waste (Management & Transboundary Movement) Rules, 2016 authorisation.
R8	**Environmental Compensation for Non-Compliance**	Liability for environmental compensation charges arises on violation of any Rule provision, e.g. shortfall in recycling targets.
R9	**Registration & Annual Maintenance Fees**	Prescribed registration fee (per recycling-target slab) and Annual Maintenance Charge (Rs. 5,000/-) must be paid on the EPR Portal.
R10	**Audit of Stakeholders**	Registered stakeholders are subject to periodic audit verifying EPR and RoHS compliance.

TABLE 2. APPLICABILITY SUMMARY BY COMPANY

Company	Primary Role	Not Applicable	Applicable (of 10)
Dixon Technologies (India) Ltd.	Manufacturer & Producer - own-brand and contract manufacturing of consumer electronics, mobiles, LED TVs, appliances	None	**10**
Kaynes Technology India Ltd.	Manufacturer / EMS - electronics system design & manufacturing, including for other brands	R8	**9**
Foxconn India (Hon Hai)	Contract Manufacturer - mobile phones, electronic components, telecom equipment for global OEM brands	R2, R3, R8	**7**
Tata Electronics Pvt. Ltd.	Manufacturer & Producer - semiconductor assembly/test, EMS, and component manufacturing	None	**10**
VVDN Technologies	Manufacturer / ODM - networking, IoT and electronics products, including some own-brand lines	R8	**9**

Note: 'Not Applicable' above reflects rules unlikely to attach to a company's role as a pure contract manufacturer (e.g., recycling-target obligations that fall on the brand-owning Producer). It is a role-based inference from public information, not a confirmed regulatory finding.

TABLE 3. RULE × COMPANY APPLICABILITY MATRIX

Rule	Dixon	Kaynes	Foxconn	Tata	VVDN
R1	✓	✓	✓	✓	✓
R2	✓	✓	—	✓	✓
R3	✓	✓	—	✓	✓
R4	✓	✓	✓	✓	✓
R5	✓	✓	✓	✓	✓
R6	✓	✓	✓	✓	✓
R7	✓	✓	✓	✓	✓
R8	✓	—	—	✓	—
R9	✓	✓	✓	✓	✓
R10	✓	✓	✓	✓	✓

✓= applicable to this company's disclosed role; — = not applicable / role-dependent (see Table 2 note).

TABLE 4. REGISTRATION & MAINTENANCE FEE STRUCTURE (CPCB ANNEXURE I)

Registration Category (per CPCB Annexure I)	Fee
Producer - Annual Recycling Target < 50 MT	**Rs. 2,500/-**
Producer - 50 MT to < 100 MT	**Rs. 7,500/-**
Producer - 100 MT to < 1000 MT	**Rs. 1,50,000/-**
Producer - 1000 MT to < 5000 MT	**Rs. 10,00,000/-**
Producer - > 5000 MT	**Rs. 15,00,000/-**

Registration Category (per CPCB Annexure I)	Fee
New Recycler (first registration, valid 5 years)	**Rs. 15,000/-**
New Refurbisher (first registration, valid 5 years)	**Rs. 15,000/-**
Manufacturer (one-time registration)	**Rs. 15,000/-**
Annual Maintenance Charge (all stakeholders)	**Rs. 5,000/-**

Producers seeking EPR registration are slabbed by annual e-waste recycling target (MT = metric tonnes); recyclers, refurbishers and manufacturers pay flat one-time/renewal fees plus an annual maintenance charge. Actual slab applicable to each of the five companies is not disclosed in public sources and is therefore not attributed here.

1. SCOPE AND BASIS OF THIS REPORT

This report identifies which provisions of the E-Waste (Management) Rules, 2022 (and amendments thereafter) apply to five companies drawn from the list of 'Top 10 Electronics Manufacturers in India': Dixon Technologies, Kaynes Technology, Foxconn India, Tata Electronics, and VVDN Technologies.

The underlying sources - the CPCB FAQ document and the attached journal article by Chadha & Chadha (2022) - are both generic regulatory/academic references and contain no company-specific compliance records. Applicability is therefore inferred from each company's publicly known manufacturing role, and the compliance-gap list in Section 4 is drawn from documented sector-wide challenges rather than company-specific audit findings.

Method: each of the ten core obligations in Table 1 was checked against the definitions of 'Manufacturer', 'Producer' and 'Refurbisher' under the 2022 Rules, and matched to each company's disclosed manufacturing model (own-brand production, contract/OEM manufacturing, or a mix of both).

2. COMPANY-WISE APPLICABILITY

2.1 Dixon Technologies (India) Ltd.

Role: Manufacturer & Producer - own-brand and contract manufacturing of consumer electronics, mobiles, LED TVs, appliances

Applicable rules: 10 of 10

- R1 - EPR Registration on the E-Waste EPR Portal
- R2 - Extended Producer Responsibility (EPR) Obligation
- R3 - Purchase of EPR Certificates from Registered Recyclers
- R4 - Reduction of Hazardous Substances (RoHS) Compliance
- R5 - RoHS Self-Declaration & Technical Documentation
- R6 - Quarterly and Annual Returns
- R7 - Consent to Operate (CTO) / Hazardous Waste Authorisation
- R8 - Environmental Compensation for Non-Compliance
- R9 - Registration & Annual Maintenance Fees
- R10 - Audit of Stakeholders

As an own-brand producer and large-scale EMS manufacturer, Dixon carries both Manufacturer and Producer obligations - the full set of 10 provisions applies.

2.2 Kaynes Technology India Ltd.

Role: Manufacturer / EMS - electronics system design & manufacturing, including for other brands

Applicable rules: 9 of 10

- R1 - EPR Registration on the E-Waste EPR Portal
- R2 - Extended Producer Responsibility (EPR) Obligation
- R3 - Purchase of EPR Certificates from Registered Recyclers
- R4 - Reduction of Hazardous Substances (RoHS) Compliance
- R5 - RoHS Self-Declaration & Technical Documentation
- R6 - Quarterly and Annual Returns
- R7 - Consent to Operate (CTO) / Hazardous Waste Authorisation
- R9 - Registration & Annual Maintenance Fees
- R10 - Audit of Stakeholders

Not applicable / role-dependent: R8

Kaynes primarily manufactures EEE for client/OEM brands; where it also sells under its own brand it additionally attracts full Producer EPR obligation (R2, R3). Environmental compensation (R8) applies only on default.

2.3 Foxconn India (Hon Hai)

Role: Contract Manufacturer - mobile phones, electronic components, telecom equipment for global OEM brands

Applicable rules: 7 of 10

- R1 - EPR Registration on the E-Waste EPR Portal
- R4 - Reduction of Hazardous Substances (RoHS) Compliance
- R5 - RoHS Self-Declaration & Technical Documentation
- R6 - Quarterly and Annual Returns
- R7 - Consent to Operate (CTO) / Hazardous Waste Authorisation
- R9 - Registration & Annual Maintenance Fees
- R10 - Audit of Stakeholders

Not applicable / role-dependent: R2, R3, R8

As a contract/OEM manufacturer, Foxconn's core obligations are RoHS compliance, CTO/hazardous-waste authorisation and return filing; the EPR recycling-target obligation (R2, R3) typically sits with the brand owner (Producer).

2.4 Tata Electronics Pvt. Ltd.

Role: Manufacturer & Producer - semiconductor assembly/test, EMS, and component manufacturing

Applicable rules: 10 of 10

- R1 - EPR Registration on the E-Waste EPR Portal
- R2 - Extended Producer Responsibility (EPR) Obligation
- R3 - Purchase of EPR Certificates from Registered Recyclers
- R4 - Reduction of Hazardous Substances (RoHS) Compliance
- R5 - RoHS Self-Declaration & Technical Documentation
- R6 - Quarterly and Annual Returns
- R7 - Consent to Operate (CTO) / Hazardous Waste Authorisation
- R8 - Environmental Compensation for Non-Compliance
- R9 - Registration & Annual Maintenance Fees
- R10 - Audit of Stakeholders

Operating both as a manufacturer of EEE/components and, through Tata Group brands, as a producer, the company is expected to fulfil the complete set of obligations.

2.5 VVDN Technologies

Role: Manufacturer / ODM - networking, IoT and electronics products, including some own-brand lines

Applicable rules: 9 of 10

- R1 - EPR Registration on the E-Waste EPR Portal
- R2 - Extended Producer Responsibility (EPR) Obligation
- R3 - Purchase of EPR Certificates from Registered Recyclers
- R4 - Reduction of Hazardous Substances (RoHS) Compliance
- R5 - RoHS Self-Declaration & Technical Documentation
- R6 - Quarterly and Annual Returns
- R7 - Consent to Operate (CTO) / Hazardous Waste Authorisation
- R9 - Registration & Annual Maintenance Fees
- R10 - Audit of Stakeholders

Not applicable / role-dependent: R8

VVDN's ODM manufacturing draws RoHS, registration, return-filing and CTO obligations; EPR recycling-target/certificate obligations (R2, R3) apply fully where it sells under its own brand.

3. E-WASTE GENERATION STATISTICS

Understanding the scale of e-waste generation - globally and in India - contextualises why the ten obligations in Table 1 exist and why enforcement capacity matters as much as the rule text itself.

3.1 Global and National Generation Trends

- Global e-waste generation reached 53.6 Mt in 2019 (7.3 kg per capita), up 9.2 Mt since 2014, and is projected to reach 74.7 Mt by 2030 - nearly double within 16 years.
- Asia produced the most e-waste in 2019 (24.9 Mt), followed by the Americas (13.1 Mt) and Europe (12 Mt); Africa and Oceania generated 2.9 Mt and 0.7 Mt respectively.
- India generated 3.2 million tonnes of e-waste in 2019, ranking third globally after China and the United States; CPCB separately estimated 10,14,961.2 tonnes for 21 EEE types based on 1,380 producers' sales data for 2019-20.
- Only 9.3 Mt (17.4%) of global e-waste generated in 2019 was documented as formally collected and recycled; Europe led at 42.5% collection/recycling, Asia followed at 11.7%, Americas and Oceania at 9.4% and 8.8%, and Africa lowest at 0.9%.
- In India specifically, only about one-fourth of the 3.2 Mt generated in 2019 could be recycled domestically.

3.2 Year-wise E-Waste Generation in India (Million Metric Tonnes)

Year	E-waste (Mt)
2015	**1.97**
2016	**2.22**
2017	**2.53**
2018	**2.86**
2019	**3.23**

Source: Centre for Science and Environment (CSE), 2020, compiled from the Global E-waste Statistics Partnership - as cited in the attached journal article. Generation nearly doubled between 2015 and 2019, underscoring the rising base against which the five companies' EPR obligations (R2, R3) are calculated.

4. HEALTH AND ENVIRONMENTAL IMPACT OF HAZARDOUS SUBSTANCES

The RoHS obligation (R4/R5 in Table 1) exists because of well-documented health and ecological harm from hazardous constituents in e-waste. The table below summarises the main chemical groups discussed in the source article.

Substance Group	Found In	Health Concern
Lead, Cadmium, Mercury, Hexavalent Chromium	CRTs, printed circuit boards, batteries, switches, relays	Persistent, bio-accumulative, toxic; kidney damage, adverse neurodevelopment, cardiovascular and respiratory effects
Brominated Flame Retardants (BFRs), PBB, PBDE	Plastic housings, circuit boards	Adverse birth outcomes, altered neuro-development, hormonal disruption
Beryllium, Antimony, Arsenic, Barium	Motherboards, finger clips, transistors	Skin disorders, immune-system effects, carcinogenic potential
CFCs (Chloro-fluorocarbons)	Fridges, coolers	Ozone-layer depletion; adverse respiratory effects on exposure
Dioxins and Furans (secondary emissions)	Generated during incomplete/informal recycling	Highly toxic combustion by-products; long-range environmental persistence

Impacts occur through three emission pathways: primary emissions from heavy metals in intact e-waste, secondary

emissions (dioxins/furans) from incomplete treatment, and tertiary emissions from the chemicals used in informal recycling itself. Informal-sector recycling - the sector-wide gap discussed in Section 6 - is disproportionately associated with these secondary and tertiary pathways.

5. INTERNATIONAL AND HISTORICAL LEGAL FRAMEWORK

The 2022 Rules did not emerge in isolation. The source article situates India's e-waste law within a wider global treaty framework and a three-quarter-century domestic legislative history, both summarised below for reference.

5.1 Global Treaty Framework

The Basel Convention (1989/1992)

The Basel Convention on the Control of Trans-boundary Movements of Hazardous Wastes and their Disposal, signed by 187 countries, is the foundational global treaty restricting cross-border hazardous-waste (including e-waste) movement; the 2006 Conference of the Parties produced the Nairobi Declaration on environmentally sound e-waste management.

The Bamako and Rotterdam Conventions

The Bamako Convention (1991/1998) bans hazardous-waste imports into Africa with no exceptions even for radioactive materials. The Rotterdam Convention regulates trade in hazardous chemicals and pesticides through Prior Informed Consent; India acceded on 24 May 2005.

The WEEE Directive, EU (2012/19/EU)

Implementing the Basel Convention within the European Union, the Waste Electrical and Electronic Equipment Directive prohibits exports of hazardous waste from EU member states to developing countries, and now regulates the majority of e-waste across the EU and Norway.

The RoHS Directive, EU (2002/95/EC; RoHS 3, 2015/863)

Adopted in February 2003, the Restriction of Hazardous Substances Directive constrained six hazardous constituents in EEE manufacturing; RoHS 3 (in force as of the source article) added four further substances. India's Rule 16(1) concentration limits mirror this approach.

Import bans and second-hand product rules in India

Under the Hazardous and Other Wastes (Management and Trans-boundary Movement) Rules, 2016, importation of e-waste from developed countries is banned in India; only refurbishing of second-hand products is permitted, and such products must be re-exported to their country of origin within one to three years.

5.2 Evolution of Indian E-Waste Law

The current Rules are the product of a stepwise legislative history:

- 1948 - Factories Act, 1948 - lists contaminants arising from manufacture or recycling of electronic components.
- 1986 - Environmental Protection Rules, 1986 - electro-plating category PCB units brought under effluent-disposal norms (Schedule 1).
- 1989/2003 - Hazardous Waste (Management and Handling) Rules, 1989 (amended 2003) - Schedules 2 & 3 applied to e-waste disposal and recycling.
- 2011/2016 - E-Waste (Management and Handling) Rules, 2011 (superseded 2016) - introduced Extended Producer Responsibility, excluding the informal sector; made recycler/dismantler registration with SPCBs obligatory.
- 2016 - Parliamentary Committee (Subordinate Legislation) 15th Report to the Lok Sabha (10 Aug 2016) - recommended a robust enforcement mechanism and adequate recycling infrastructure.
- 2018 - E-Waste (Management) Amendment Rules, 2018 - introduced channelling of e-waste to authorised dismantlers/recyclers to formalise the sector.
- 2022/2023 - E-Waste (Management) Rules, 2022, effective 1 April 2023 - current EPR-certificate framework, RoHS Rule 16(1), and 106-EEE, seven-category Schedule-I now governing the five companies profiled in this report.

6. SECTOR-WIDE COMPLIANCE GAPS (NOT COMPANY-SPECIFIC)

The source paper documents several structural gaps in India's e-waste regulatory ecosystem. These are systemic risk areas that affect the entire Indian EEE-producer base - including, but not attributable specifically to, the five companies profiled above - rather than confirmed violations by any named company.

6.1 Heavy reliance on the informal/unorganized sector

The paper notes most collection, transportation and recycling in India is handled by a well-networked but unregulated informal sector, so producer-financed collection frequently leaks into unauthorised channels rather than registered recyclers.

6.2 Opaque public data on recyclers and collection centres

“On most State Pollution Control Board (SPCB) websites, there is absence of public information especially regarding details of recyclers and collectors of e-waste. Even the basic e-waste Rules and guidelines have not been uploaded,” limiting independent verification of any producer's downstream compliance chain.

6.3 Collection vs. generation gap

Only about one-fourth of India's e-waste generated (2019) was actually recycled through formal channels - a structural shortfall that affects the whole producer base, not one firm.

6.4 Weak penal enforcement on cross-border e-waste/second-hand flows

The paper records that customs and environment officials lack adequate devices to distinguish second-hand products from e-waste, and “the law is silent on the penal clause” even where fraud is detected - a systemic enforcement gap.

6.5 Limited use of LCA / MFA / MCA in EPR design

The study recommends Life Cycle Assessment, Material Flow Analysis and Multi-Criteria Analysis to strengthen EPR policy - tools not yet mandated for producers under the current Rules.

6.6 No consumer deposit-refund or take-back incentive mechanism

The paper recommends deposit mechanisms and separate e-waste bins with penalties, modelled on Japan - features presently absent from the Indian regulatory design, so no producer is currently required to offer them.

7. RECOMMENDATIONS

Drawing on the source paper's own findings, the following measures would strengthen compliance verification across the sector - and could reasonably be adopted voluntarily by any of the five companies ahead of stricter enforcement:

- Integrate Life Cycle Assessment (LCA), Material Flow Analysis (MFA) and Multi-Criteria Analysis (MCA) into EPR compliance reporting, not only recycling-target arithmetic.
- Require Producer Responsibility Organisations (PROs) - shared producer/industry bodies - to formally enforce take-back, on the lines of the Norwegian system, tailored to Indian conditions.
- Introduce consumer deposit-refund mechanisms and mandated separate e-waste bins with penalties for non-use, drawing on the Japanese model referenced in the source literature.
- Publish recycler and collection-centre details on every SPCB website, closing the transparency gap the source paper identifies.
- Track components (circuit boards, refrigerant compressors) with radio-frequency identification tags to support a global monitoring system across the product life cycle.
- Provide fiscal incentives to the formal e-waste processing industry and a disposal fee (2-5% of original production cost) payable by the country ultimately responsible for disposal, to curb illegal cross-border trade.

8. LIMITATIONS OF THIS REPORT

This report is a desk-based regulatory-applicability mapping exercise. Readers should note the following constraints before treating any statement above as a verified compliance finding:

- Company applicability (Table 2/3) is inferred from publicly known manufacturing roles, not from each company's own EPR Portal registration filing, which is not public.
- The report does not have access to company-specific CPCB/SPCB audit reports, environmental compensation orders, or BRSR/ESG disclosures that would confirm actual (as opposed to expected) compliance.
- Generation and recycling statistics cited (Section 3) are national/global aggregates and cannot be disaggregated to any single producer's contribution.
- The fee slab each company falls into (Table 4) depends on its actual annual recycling target/sales volume, which is confidential business information not used in this report.

9. CONCLUSION

All five companies, as manufacturers of Electrical and Electronic Equipment listed in Schedule-I, fall within the applicability of the E-Waste (Management) Rules, 2022. Companies that also sell under their own brand (Dixon, Tata Electronics, and partly Kaynes/VVDN) carry the full EPR recycling-target and certificate-purchase obligations, while a pure contract manufacturer such as Foxconn India primarily bears RoHS, registration and reporting obligations, with EPR recycling targets generally resting with the brand-owning Producer.

Whether any individual company is meeting these obligations in practice cannot be established from the two source documents used here, since neither contains firm-level compliance data; verification would require each company's own EPR Portal filings, CPCB/SPCB audit records, or annual BRSR/sustainability disclosures.

Read together with Section 3, the picture is one of a comprehensive rule-book applied against a still-maturing enforcement infrastructure - the obligations on paper (Table 1) are considerably more complete than the transparency and enforcement mechanisms (Section 4) needed to verify them in practice.

REFERENCES

1. Central Pollution Control Board (CPCB), WM-III Division. Frequently Asked Questions (FAQ) under E-Waste (Management) Rules, 2022. 23 January 2024.

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3. E-Waste (Management) Rules, 2022, Ministry of Environment, Forest and Climate Change, Government of India (effective 1 April 2023).

4. Hazardous and Other Wastes (Management and Trans-boundary Movement) Rules, 2016, Government of India.

5. MoKo Technology. 'List of Top 10 Electronics Manufacturers in India.' (source of company shortlist used in this report).

Additional primary-source citations (Basel, Bamako and Rotterdam Conventions; EU WEEE and RoHS Directives; Indian legislative history 1948-2018) follow the numbered references 1-36 of source reference 2 above and are not reproduced verbatim here; consult the original article for full citation text and URLs.